Gas, coal or oil? The cheapest fuel for power in Germany and Great Britain

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Question: predict when it is cheaper to use natural gas, heating oil or coal for power generation, using clean spark and dark spreads, for Europe and the UK.

1. Data and method

Daily Bloomberg front-month futures (`xbbg`, `PX_LAST`), 2 Jan 2019 to 9 Nov 2026 (2,048 trading days): German (DE) and British (GB) baseload power, TTF and NBP gas, API2 coal, ICE gasoil (heating-oil proxy), EUA and UKA carbon (plus the £18/t Carbon Price Support in GB), all in euros. Clean spread: $CS = P - (F + EF \cdot C)/\eta$, with $\eta = 51\%$ (gas), 35% (coal, oil) and $EF = 0.202$, 0.341, 0.267 tCO_2/MWh of fuel. Plants share one power price, so **the highest clean spread marks the cheapest fuel**.

2. When is each fuel cheapest?

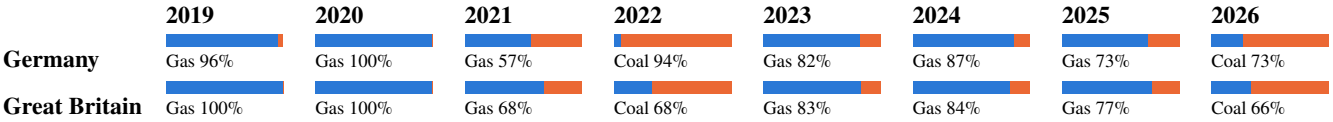


Figure 1: Cheapest fuel, share of trading days per year (gas, coal; oil never ranks first). 2026 runs to 9 November.

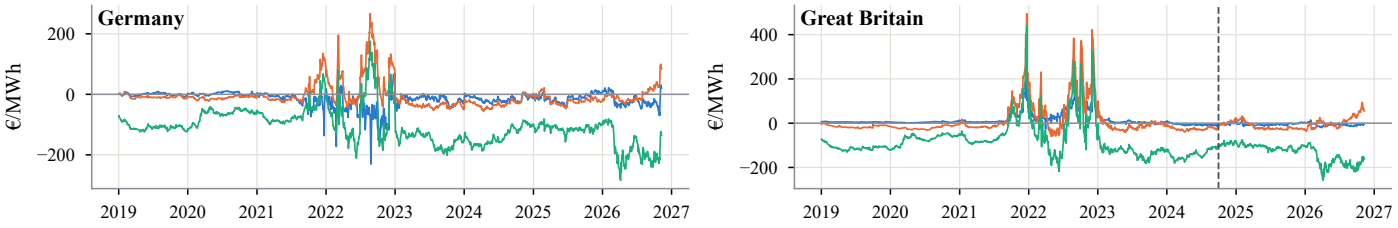


Figure 2: Daily clean spreads, €/MWh: gas (spark), coal (dark), oil (gasoil); dashed line: last GB coal plant closes. On any day, the highest line is the cheapest fuel.

- **Oil never wins**: its spread sits a median €106 (DE) and €111/MWh (GB) below the best fuel, as gasoil is costly and oil plants are inefficient.
- **Gas is the default**: cheapest on 67% of days in Germany and 73% in Britain, including all of 2019–20 and most of 2023–25.
- **Coal wins when gas is scarce**: August 2021 to January 2023 (Russian supply cuts, TTF peak of €339/MWh in August 2022), winter 2024–25, and again since July 2026.
- **One rule explains the switch**: gas beats coal when carbon trades above $C^* = (P_{gas}/0.51 - P_{coal}/0.35)/0.578$, which matches the dominant fuel every year in both markets (2022: $C^* =$ €213/t vs EUA €82/t). The £18/t GB carbon tax is why coal never won there in 2019–20.

3. Can we predict tomorrow's cheapest fuel? An AR(1) model

Each spread is regressed on its previous value, $CS_{t+1} = c + \phi\, CS_t + \varepsilon_{t+1}$ (OLS), refitted every month from January 2022 on past data only (walk-forward, 1,264 test days), and compared with the naive forecast $CS_{t+1} = CS_t$.

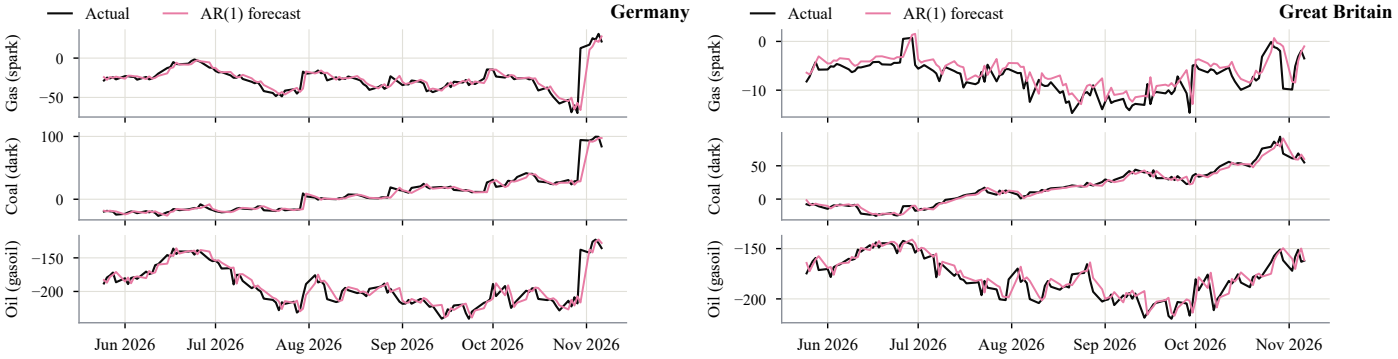


Figure 3: Actual and AR(1) forecast of next-day clean spreads over the last 120 test days, €/MWh.

	Germany			Great Britain		
	ϕ	AR(1)	Naive	ϕ	AR(1)	Naive
Gas	0.93	10.90	11.08	0.93	15.18	15.42
Coal	0.98	9.62	9.61	0.97	17.20	17.31
Oil	0.99	12.00	12.01	0.98	18.31	18.39
Best fuel	96.8% of days			95.3% of days		

Table 1: Out-of-sample RMSE, €/MWh. ϕ: full sample.

With $\phi = 0.93$–0.99 the spreads are close to a random walk. AR(1) beats the naive forecast only for gas (−1.6% RMSE in DE, −1.5% in GB: slight mean reversion). It ranks the right fuel first on 96.8% (DE) and 95.3% (GB) of days, vs 96.8% and 95.6% for keeping today's leader. The big misses (Fig. 3, late October) are contract rolls: the 4.6% of days before a month turn carry 18% (DE) and 53% (GB) of the naive squared error.

4. Conclusion

Gas is the cheapest fuel whenever carbon trades above the gas–coal switching price, as in most of 2019–2026; coal takes over only when gas is scarce, and oil is never competitive. One day ahead, today's ranking is the best forecast: an AR(1) with $\phi \approx 1$ adds little, and predicting switches needs data that front-month prices lack (forward curve, storage, weather). On 9 Nov 2026 coal leads in both markets; German gas would need TTF near €50/MWh (now €82.5) or EUA near €197/t (now €88).

Limits: one efficiency per technology, month-ahead baseload prices rather than hourly dispatch, no switching costs; British coal is theoretical since Ratcliffe-on-Soar closed (30 Sep 2024).